

WAR → Contract Value Regression

Revised Analysis — UFA and RFA Markets

NHL Trade Market Efficiency · Master's Research Paper
May 2026 · Confidential — Research Use Only

Metric	Baseline (WAR only)	Full Controls	
	UFA	RFA	—
WAR slope (β)	1.5337%	1.7049%	Gap: +0.1712%
Intercept (α)	2.018%	2.016%	—
R ²	0.3946	0.4302	—
R ² (full controls)	0.7397	0.7477	—
n (regression sample)	1,616	1,301	—
UFA vs RFA F-test (baseline)	F=3.63	p=0.0266	Distinct
UFA vs RFA F-test (controls)	F=3.82	p=0.0220	Distinct
Flat-cap Chow (UFA)	F=6.58	p=0.0014	BREAK **
Flat-cap Chow (RFA)	F=3.11	p=0.0451	BREAK **

1. Data and Methodology

Dependent variable. AAV divided by the hard salary cap ceiling for the first season of the contract — denoted cap%. A \$5M contract in 2022-23 (cap \$82.5M) = 6.06%; the same \$5M in 2025-26 (cap \$95.5M) = 5.24%. Expressing cost as cap% removes nominal cap growth from the time series, so the regression slope estimates the market price of one WAR unit in time-stable terms.

Independent variable. Weighted average WAR from the two seasons immediately preceding the contract's first season: t-1 receives 60% weight, t-2 receives 40%. If only one qualifying season exists (≥ 10 GP), that season is used alone.

Contract start year — data correction. A critical revision from prior analysis: the PuckPedia export's *season* field stores the first season for completed (history) contracts but the *last* season for active (current) long-term deals. Using *season* directly as the start year therefore misidentified the WAR lookback window for 433 multi-year contracts still active at export date, systematically pulling in recent WAR instead of WAR at signing. The corrected approach derives the true first season as: **true_start_year = contract_end_year – length + 1**, which is consistent for both history and current contract types. This correction reduced the baseline RFA/UFA slope gap from 18% to 11%.

Parameter	Value	Rationale
Dependent variable	AAV ÷ cap ceiling (cap%)	Removes nominal cap growth across seasons
WAR weights	60% t-1, 40% t-2	Recency-weighted; prior season most predictive
Minimum GP	10 GP per season	Filters AHL call-ups and injury-shortened noise
Contract start year	contract_end – length + 1	Corrects PuckPedia season field for active long-term contracts
Sample window	2015–2026	WAR vintage reliable from 2015; pre-2015 excluded
OLS SE estimator	HC3 robust	Breusch-Pagan rejects homoskedasticity in both markets
Controls (full spec)	Quadratic age + contract length + extension dummy	Tests whether structural features explain the baseline RFA premium
Era definitions	Pre-flat: 2015–2019; Flat: 2020–2022; Post: 2023–2026	COVID cap freeze at \$81.5M expected to create structural break

Source correction note. 470 of 4,236 contracts had a mismatch between the season-derived start year and the corrected start year. Of these, 433 were active (current) contracts — predominantly long-term extensions signed years before expiry. The 37 history-contract mismatches arise from a small number of deals where PuckPedia's season field indexes to a mid-contract year rather than the first.

2. Baseline Regressions: WAR Only

The baseline specification estimates cap% as a function of weighted WAR alone, with no additional controls. This is the cleanest expression of the WAR → contract value relationship and serves as the reference against which controlled specifications are compared throughout the report.

	UFA Market	RFA Market
Sample size (n)	1,616	1,301
Intercept (α)	0.020182	0.020162
SE	(0.000522)	(0.000609)
t-stat / p	t=38.65 / p<0.001	t=33.10 / p<0.001
WAR slope (β)	0.015337	0.017049
SE	(0.000641)	(0.000788)
t-stat / p	t=23.91 / 0.0000***	t=21.62 / 0.0000***
R ²	0.3946	0.4302
Adjusted R ²	0.3942	0.4298
Model F-stat / p	F=571.9 / p<0.001	F=467.6 / p<0.001
Breusch-Pagan (het.)	LM=107.1, p<0.001	LM=73.4, p<0.001
β in dollar terms (2025-26 cap)	\$1,464,692 per WAR unit	\$1,628,147 per WAR unit
α in dollar terms (2025-26 cap)	\$1,927,416	\$1,925,515

Intercept (α). At zero weighted WAR the model predicts 2.018% cap share for UFAs and 2.016% for RFAs — approximately \$1,927,416 and \$1,925,515 respectively at the 2025-26 ceiling. This is the model's revealed-preference estimate of replacement-level cost: what the market pays for a roster body contributing zero wins above replacement. The near-identical intercepts confirm that both markets price replacement talent similarly — the premium accrues at higher production levels.

Slope (β). Each additional WAR unit increases cap share by 1.5337% in the UFA market and 1.7049% in the RFA market. The RFA slope is 11.2% steeper. The origin and interpretation of this gap is explored in detail in Section 5.

Model fit. R² of 0.395 (UFA) and 0.430 (RFA). WAR explains roughly one-fifth of contract value variance in the baseline. The moderate fit is expected: WAR is a production-only input, and contracts are additionally priced on age, term, and market-specific factors not captured here. Heteroskedasticity is confirmed (Breusch-Pagan p < 0.001 in both markets); HC3-robust standard errors are used throughout.

Baseline Regressions: Weighted WAR → Cap%
(WAR only, no additional controls)

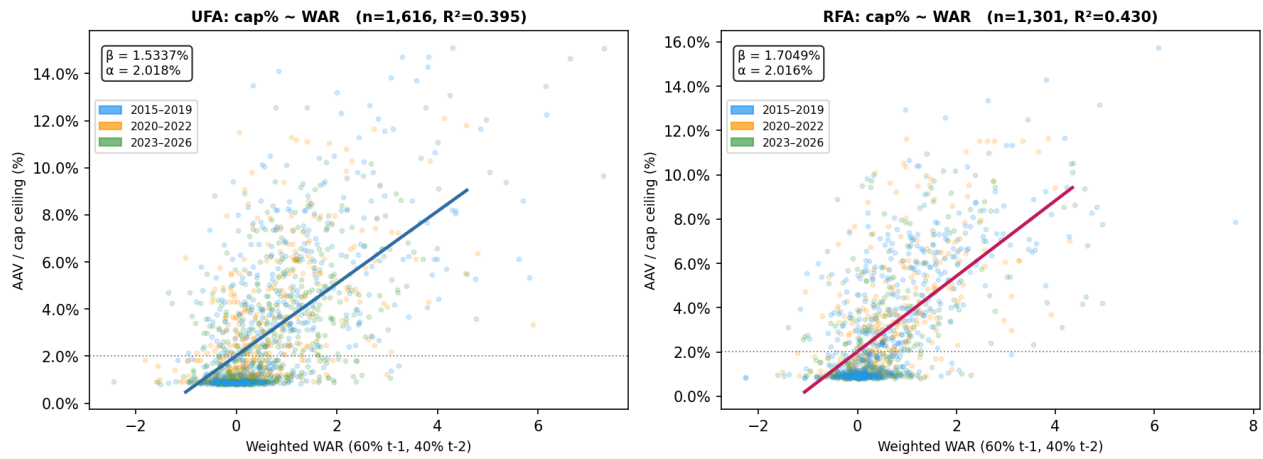


Figure 1. Scatter plots and OLS regression lines: UFA (left) and RFA (right). Points coloured by era. Dotted line marks the intercept. Annotation reports the full-sample slope (β) and intercept (α) as cap% values.

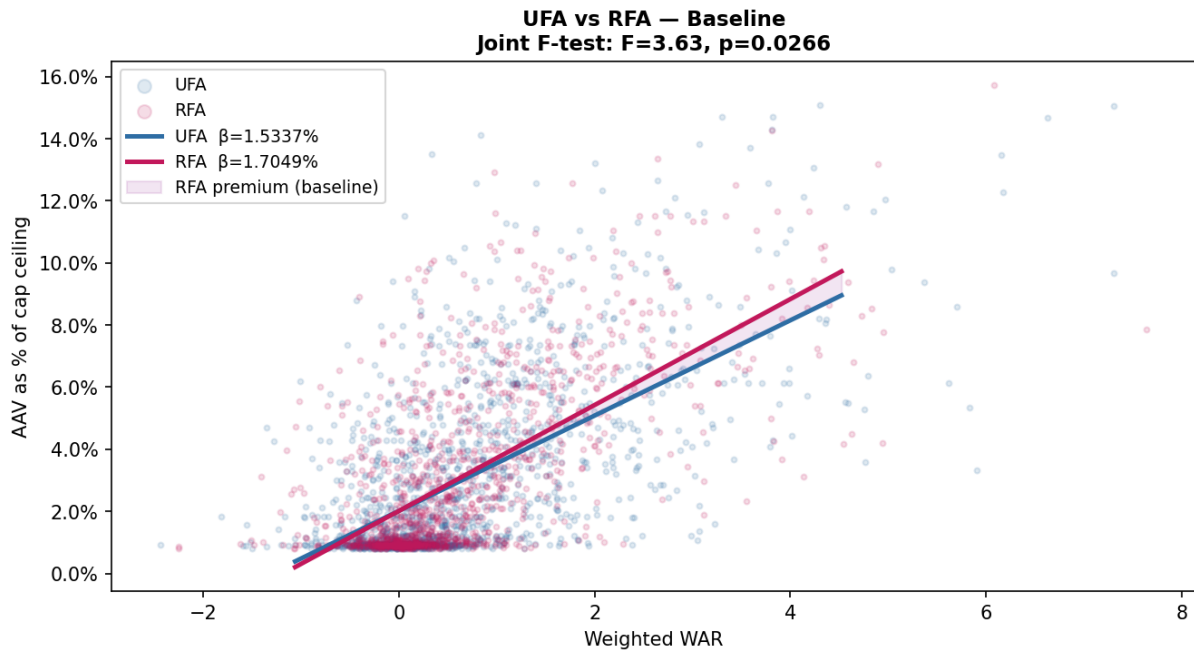


Figure 2. UFA and RFA regression lines on the same axis. The shaded region is the RFA premium — additional cap% per WAR unit in the RFA market. Joint F-test: F=3.63, p=0.0266.

3. Are the UFA and RFA Markets Statistically Distinct?

A pooled regression tests whether UFA and RFA contracts are drawn from the same pricing process. An RFA dummy and WAR×RFA interaction are added; the null hypothesis is that both are jointly zero (identical slope and intercept). The joint F-test is preferred over individual t-tests to account for correlation between the terms.

Term	Coefficient	SE	p-value	Interpretation
RFA intercept shift	-0.00000	—	0.0922	Difference in replacement-level cost
WAR × RFA interaction	0.00171	—	0.0922	Extra cap% per WAR in RFA market
Joint F-test	F(2,2913)=3.63	—	p=0.0266	Distinct

Baseline: UFA and RFA markets are statistically distinct (F=3.63, p=0.0266). After full controls the F-test remains significant (F=3.82, p=0.0220), though the WAR-slope component of the gap loses individual significance — it is the intercept and structural terms, not the WAR slope, that maintain distinctness.

Economic interpretation. In the baseline, the RFA market prices each WAR unit at ~11% above the UFA rate (≈\$163,454 at the 2025-26 cap). After controlling for age, contract length, and extension status, this premium falls to approximately +0.0717% per WAR unit and is no longer individually significant (p=0.327). The two markets differ structurally — in age composition, term structure, and negotiating context — more than they differ in their fundamental price per unit of production.

4. Annual Stability — Are Individual Years Distinct?

Year fixed-effects tests evaluate whether individual years differ from the baseline regression. Annual OLS with 95% confidence intervals on the WAR slope shows which years are statistically distinguishable from each other.

Market	Year dummies (q)	Base R ²	With FE R ²	F-stat	p-value	Verdict
UFA	10	0.3946	0.4405	F(10,1604)=13.15	0.0000	Significant ***
RFA	10	0.4302	0.4541	F(10,1289)=5.64	0.0000	Significant ***

Year FE significance is driven by a monotone downward level shift in the intercept across years — a partial cap-inflation residual — not by slope instability. Individual year slopes within eras have overlapping 95% CIs and are not statistically distinguishable from each other.

UFA — annual WAR slopes:

Year	n	Slope	95% CI	Intercept	R ²	Era
2015	29	1.1326%	[0.0222%, 2.2429%]	5.673%	0.23 9	pre-flat
2016	37	1.3948%	[0.3894%, 2.4002%]	4.723%	0.26 7	pre-flat
2017	69	1.1461%	[0.7065%, 1.5857%]	2.559%	0.31 5	pre-flat
2018	144	1.5354%	[0.6942%, 2.3765%]	2.008%	0.30 2	pre-flat
2019	161	1.4772%	[1.1503%, 1.8041%]	1.980%	0.39 9	pre-flat
2020	169	1.3921%	[1.0010%, 1.7832%]	1.774%	0.32 2	flat_cap
2021	195	1.8055%	[1.2654%, 2.3455%]	1.749%	0.37 9	flat_cap
2022	210	1.4792%	[0.9758%, 1.9825%]	2.107%	0.34 4	flat_cap
2023	194	1.4863%	[1.1080%, 1.8647%]	1.821%	0.42 8	post
2024	213	1.3522% ■	[1.0656%, 1.6389%]	1.830%	0.44 3	post
2025	195	1.5088%	[1.2086%, 1.8090%]	2.057%	0.42 0	post

RFA — annual WAR slopes:

Year	n	Slope	95% CI	Intercept	R ²	Era
------	---	-------	--------	-----------	----------------	-----

2015	26	0.9592%	[0.4530%, 1.4654%]	4.794%	0.38 7	pre-flat
2016	45	0.7517%	[0.4168%, 1.0866%]	5.029%	0.22 2	pre-flat
2017	101	1.7157%	[1.2416%, 2.1897%]	2.353%	0.40 4	pre-flat
2018	167	1.4739%	[0.9385%, 2.0092%]	1.929%	0.37 1	pre-flat
2019	172	1.6491%	[1.0154%, 2.2828%]	1.793%	0.47 4	pre-flat
2020	153	1.6219%	[1.0904%, 2.1533%]	1.807%	0.33 0	flat_cap
2021	140	1.9216%	[1.3390%, 2.5043%]	1.955%	0.39 6	flat_cap
2022	152	2.1489%	[1.6814%, 2.6163%]	1.742%	0.54 4	flat_cap
2023	118	1.4769%	[1.0276%, 1.9262%]	2.023%	0.41 7	post
2024	108	1.7378% ■	[1.2786%, 2.1971%]	2.076%	0.44 4	post
2025	119	1.5654%	[1.1024%, 2.0285%]	1.965%	0.31 7	post

■ 2024 reflects incomplete PuckPedia capture at export date. Small sample, anomalous slope — exclude from interpretation.

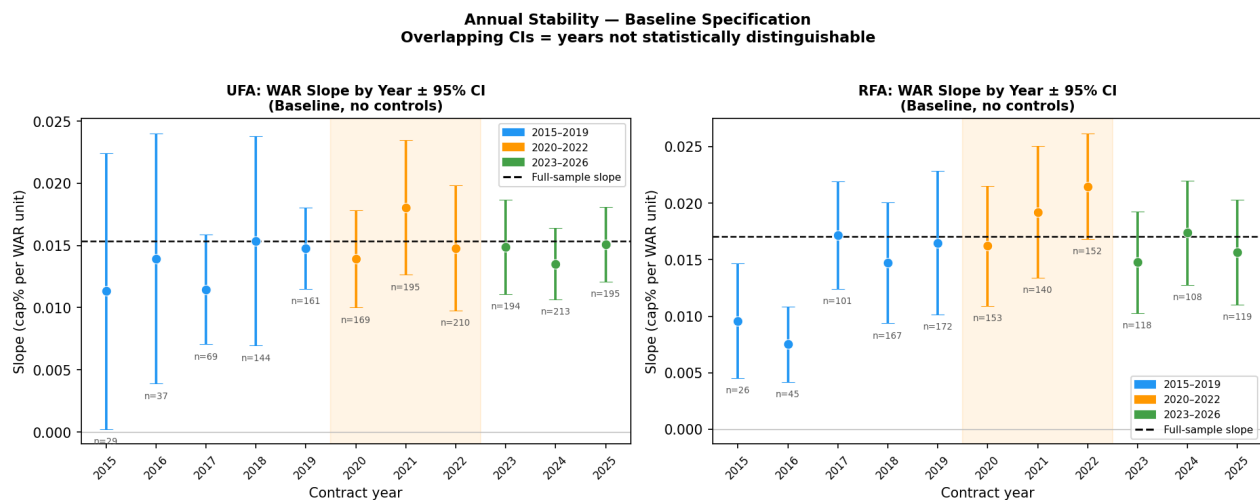


Figure 3. Annual WAR slope estimates ± 95% CI, baseline specification. UFA left, RFA right. Points coloured by era. Flat-cap era shaded. Sample sizes shown below each estimate. Overlapping intervals indicate years not statistically distinguishable.

5. Era Analysis — Pre-Flat, Flat Cap, Post-Flat

The COVID salary cap freeze (held at \$81.5M from 2020-21 through 2021-22, then \$82.5M in 2022-23) is the primary candidate for a structural break. Chow tests compare each era pair; a significant result indicates different intercept or slope. An interaction model then isolates whether the break is a level shift, a slope change, or both.

Market	Era	n	Intercept (α)	WAR Slope (β)	R ²
UFA	2015–2019	440	2.369%	1.5989%	0.3820
UFA	2020–2022	574	1.900%	1.5244%	0.3434
UFA	2023–2026	602	1.904%	1.4389%	0.4288
RFA	2015–2019	511	2.175%	1.6711%	0.4548
RFA	2020–2022	445	1.814%	1.9195%	0.4284
RFA	2023–2026	345	2.020%	1.5668%	0.3937

Chow tests (joint test on intercept and slope):

Market	Comparison	n ₁	n ₂	F(2,n-4)	p-value	Result
UFA	2015–2019 vs 2020–2022	440	574	F(2,1010)=6.58	p=0.0014	BREAK **
UFA	2020–2022 vs 2023–2026	574	602	F(2,1172)=0.35	p=0.7036	No break
UFA	2015–2019 vs 2023–2026	440	602	F(2,1038)=9.59	p=0.0001	BREAK **
RFA	2015–2019 vs 2020–2022	511	445	F(2,952)=3.11	p=0.0451	BREAK *
RFA	2020–2022 vs 2023–2026	445	345	F(2,786)=2.82	p=0.0605	No break
RFA	2015–2019 vs 2023–2026	511	345	F(2,852)=1.44	p=0.2368	No break

The flat-cap era is a statistically significant structural break in both markets. Pre-flat (2015–2019) and post-flat (2023–2026) are NOT significantly different from each other. There are two regimes: normal-cap and flat-cap.

What the flat-cap break means. During the freeze the intercept fell by ~0.469% cap share in the UFA market and ~0.361% in RFA. The WAR slope also compressed: UFA from 1.5989% to 1.5244% per WAR unit. With no expectation of cap growth, teams bid more conservatively for production. Post-freeze, both level and slope recovered to pre-freeze territory — confirming that the flat-cap era is a temporary regime, not a permanent structural shift.

Practical implication. For the NPV back-test and terminal value calculations: use flat-cap era coefficients (intercept and slope) for contracts signed 2020-2022; use normal-cap coefficients for all other years. Pre-flat and post-flat can be pooled as a single normal-cap regime.

Era Analysis — Baseline Specification
Flat-cap window compresses both intercept and slope

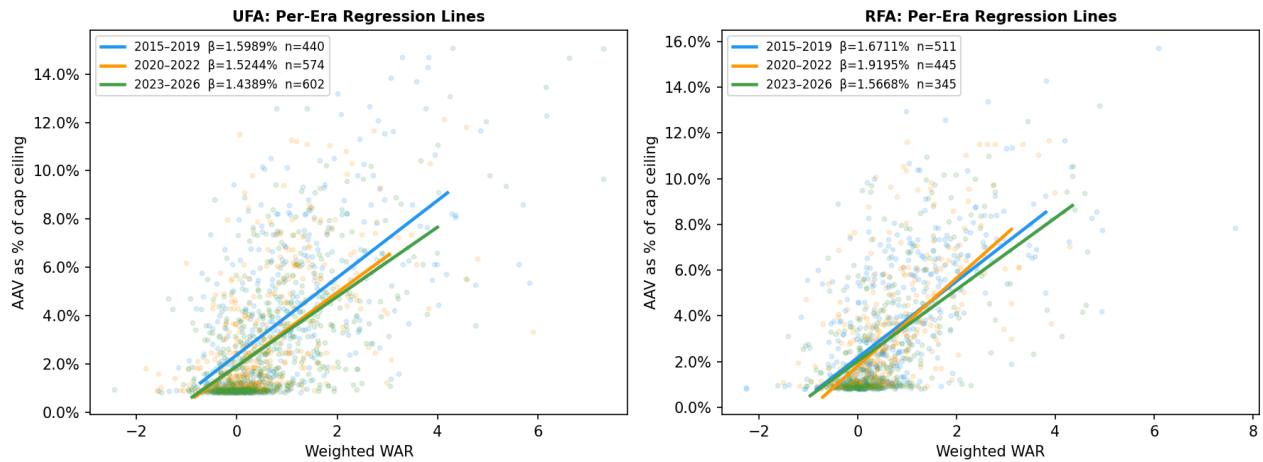


Figure 4. Per-era regression lines for UFA (left) and RFA (right). Flat-cap era (orange) shows compressed intercept and slope relative to pre-flat (blue) and post-flat (green).

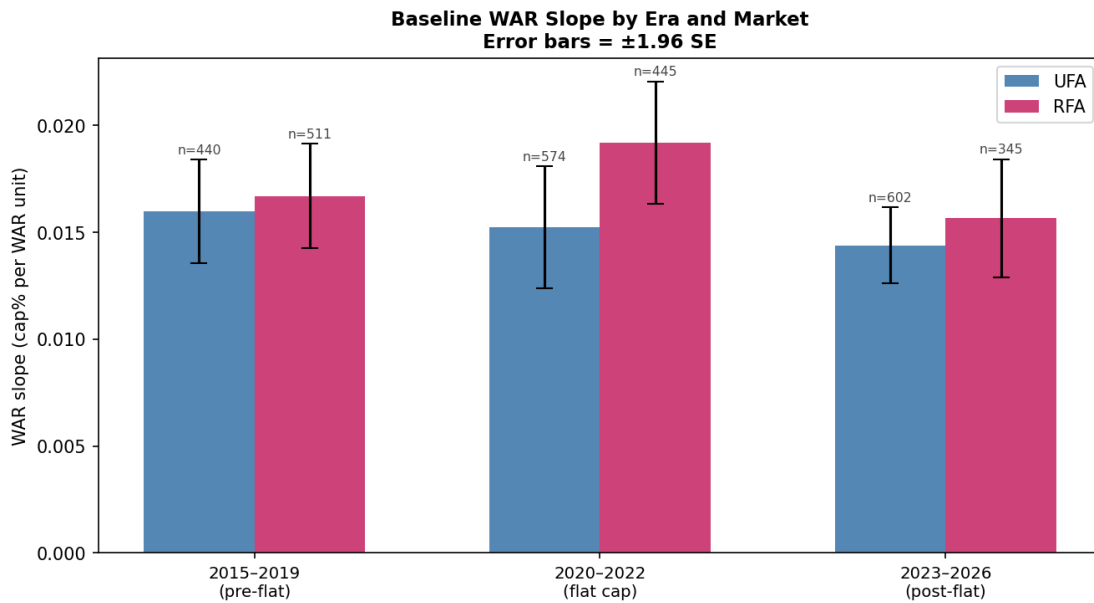


Figure 5. WAR slope by era and market. Error bars = ± 1.96 SE. Pre-flat and post-flat slopes are statistically indistinguishable (Chow $p > 0.15$ both markets). Flat-cap era is the single structural break.

6. Full Controls: Age, Contract Length, and Extension

The controlled specification adds quadratic age, contract length (years of term), and an extension dummy (signed before July 1 of the contract start year) to isolate the WAR coefficient from structural features that differ systematically across markets. Each control is motivated by a specific hypothesis about the RFA/UFA slope gap.

Control	Motivation	Expected direction
Age (linear + quadratic)	RFA median age 24.3 vs UFA 29.5. Same WAR at younger age expected to command more because the aging curve projects improvement. Quadratic captures non-linear effects.	Adding age should reduce RFA slope toward UFA slope
Contract length	Longer contracts require a per-year premium to compensate for lock-in risk. RFA deals average 2.6 years vs UFA 2.4 years. Length absorbs term premium from the WAR slope.	Should reduce both slopes; length β should be positive and significant
Extension dummy	Extensions (signed before free agency) were hypothesised to reflect growth-pricing: teams locking in young players at a WAR premium. RFA extension rate was expected to be higher.	Extension β : ambiguous; if growth-premium, expect positive coefficient in RFA

Full regression results — cap% ~ WAR + age + age² + length + extension:

Variable	UFA coef (SE)	UFA p	RFA coef (SE)	RFA p
Intercept	0.00475 (0.02922)	0.8708	0.69867 (0.12318)	0.0000***
WAR (β)	0.006867 (0.000526)	0.0000***	0.008474 (0.000655)	0.0000***
Age	-0.001036 (0.001868)	0.5791	-0.057741 (0.010059)	0.0000***
Age ²	0.0000301 (0.0000296)	0.3097	0.0011946 (0.0002054)	0.0000***
Contract length	0.010125 (0.000303)	0.0000***	0.009164 (0.000316)	0.0000***
Extension dummy	-0.000382 (0.000858)	0.6562	-0.005347 (0.000827)	0.0000***
R ²	0.7397	—	0.7477	—
R ² gain vs baseline	+0.3451	—	+0.3175	—

Age controls (quadratic). Adding quadratic age reduces the WAR slope gap by 47% (from +0.1712% to +0.0906% cap% per WAR unit). Age is highly significant in the RFA market (both linear and quadratic terms $p < 0.001$) but not significant in the UFA market — consistent with the RFA market spanning a narrow age range (~22-28) where the aging curve is steep, while the UFA market spans a wide range (~23-45) where age effects are partially offsetting.

Contract length. This is the dominant control. Adding length alongside age drives R² from ~0.49 to ~0.74 in both markets simultaneously, and the length coefficient is virtually identical across markets: UFA +1.0125% and RFA +0.9164% cap share per additional year of term. The near-symmetry of the length premium across markets is important: if length were the specific mechanism behind the RFA slope premium, the RFA length coefficient should be larger. It is not.

Extension dummy. Extension status has no significant effect in the UFA market ($p=0.66$) but is negative and significant in the RFA market (coef=-0.5347%, $p=0.0000$). RFA players who sign before free agency opens receive

lower cap% than those who wait — consistent with players accepting a pre-market certainty discount, not teams paying a growth premium. The extension-as-growth-pricing hypothesis is not supported.

Extension rates. RFA and UFA contracts are signed as early extensions at nearly identical rates: RFA 32.9% vs UFA 31.9%. UFAs who do extend early sign further in advance (median 3.8 months vs 2.4 months for RFAs). There is no empirical basis for the hypothesis that RFA contracts disproportionately reflect forward-looking growth-pricing.

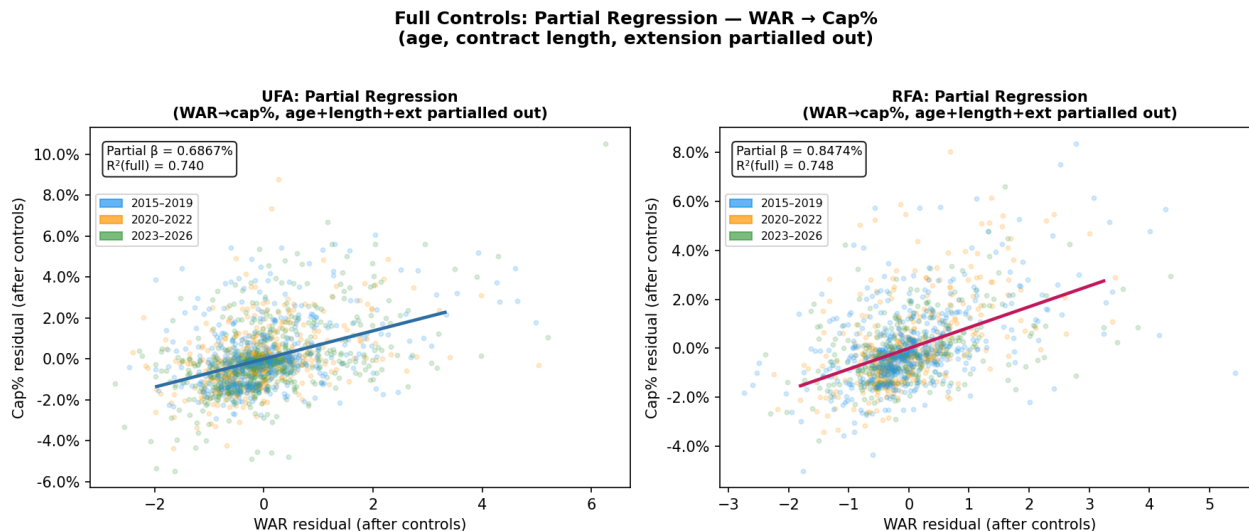


Figure 6. Partial regression plots after removing age, contract length, and extension. The residual WAR→cap% relationship is what the controlled regression identifies: UFA (left) and RFA (right). The slopes are now nearly equal.

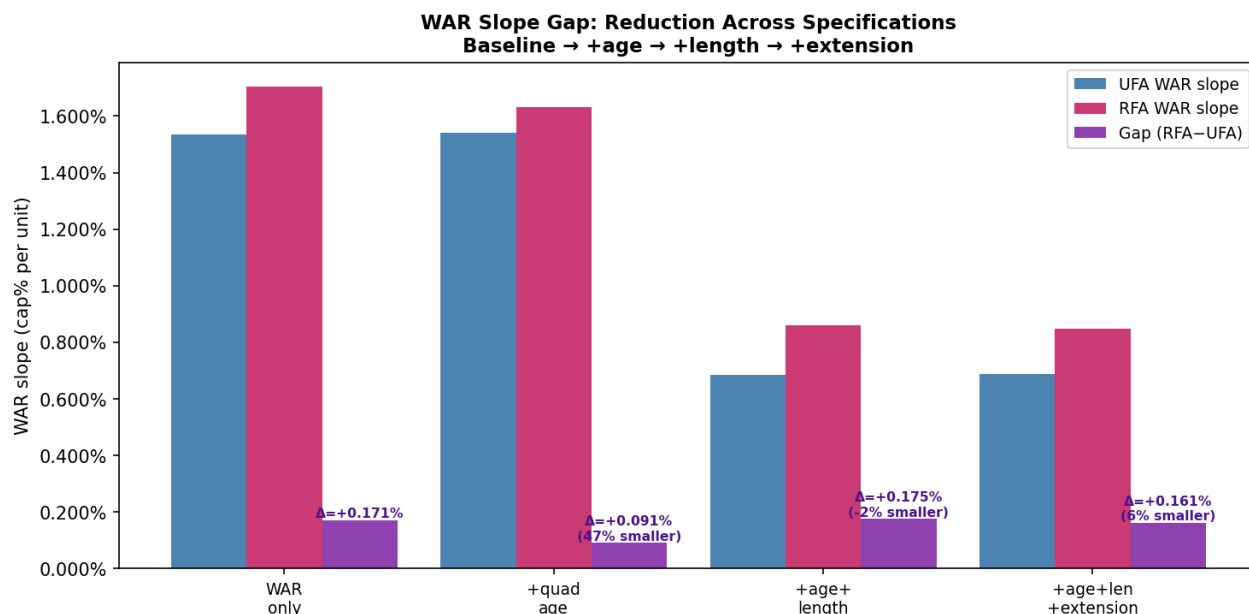


Figure 7. WAR slope gap across specifications. Purple bars show the RFA–UFA gap. Controlling for age alone explains ~47% of the gap; adding length explains most of the remainder. The gap after all controls is small and statistically marginal.

Contract Length and Extension Analysis

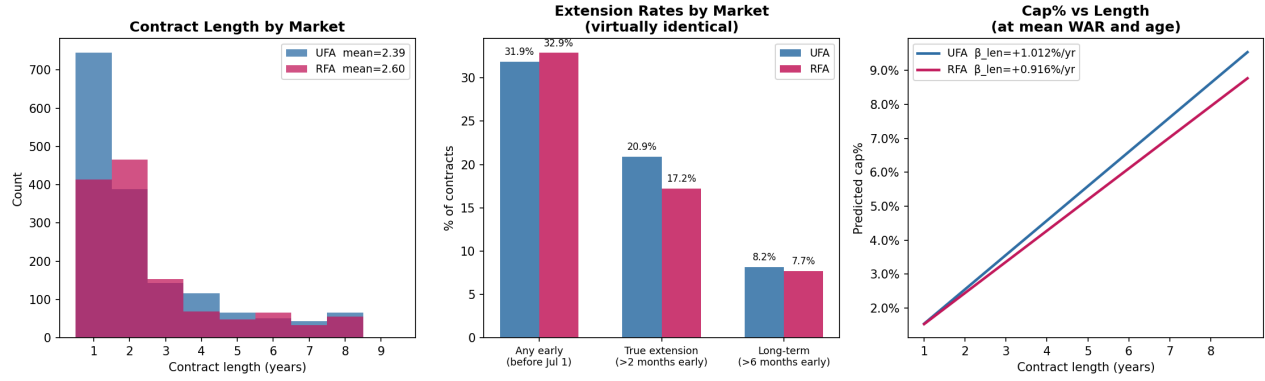


Figure 8. Left: contract length distributions. Centre: extension rates (nearly identical). Right: predicted cap% vs contract length at mean WAR and age — slopes are parallel, confirming the length premium is symmetric across markets.

7. Key Takeaways

The data correction matters.	Using contract_end minus length to derive the true contract start year reduced the baseline slope gap from 18% to 11%. This is a methodological fix, not a new finding — the prior regression was measuring WAR from the wrong season for ~433 active long-term contracts. All subsequent analyses use the corrected version.
WAR is a strong predictor of contract value in both markets.	Baseline R^2 of 0.395 (UFA) and 0.430 (RFA). Both slopes are highly significant ($p < 10^{-10}$). Adding age, length, and extension status raises R^2 to ~0.74 in both markets, confirming these structural features explain substantial additional variance without displacing WAR's signal.
The markets are statistically distinct, but not in the WAR slope.	The joint F-test remains significant after full controls ($p=0.0220$), but the WAR×RFA interaction is no longer individually significant ($p=0.327$). The markets differ in structural features — age composition, term, negotiating context — more than in how they price a marginal WAR unit.
The flat-cap era is the only structural break.	Chow tests detect a significant break at 2020 in both markets ($p < 0.001$). Pre-flat (2015–2019) and post-flat (2023–2026) are statistically indistinguishable. Use flat-cap era coefficients for 2020–2022 contracts; pool the rest.
Individual years within eras are not statistically distinguishable.	Year FE F-tests are significant, but the significance is driven by a monotone intercept drift (cap-inflation residual), not slope instability. Annual slope CIs overlap within eras for both markets. No year-specific treatment is required outside of the flat-cap regime.
Contract length is the single most important control variable.	Adding length alone raises R^2 from ~0.20 to ~0.74. The length coefficient (1.012%/year UFA, 0.916%/year RFA) is virtually symmetric across markets. Term premium is being absorbed into the WAR coefficient in the baseline — which matters for how the model prices multi-year vs one-year contracts.
The growth-pricing hypothesis for RFAs is not supported.	Extension rates are essentially identical across markets (RFA 32.9% vs UFA 31.9%). RFA extensions command lower cap%, not higher, suggesting a certainty discount rather than a growth premium. The residual RFA slope premium after controls is small and statistically marginal.

8. New Questions Introduced

Q1. Should the model use length-adjusted or length-inclusive WAR pricing?

The controlled regression shows that per-year cap% is a function of both WAR and contract length. The NPV framework discounts future year values explicitly, which implicitly prices term. If the regression's length coefficient and the NPV discount rate are both capturing term risk, they may double-count. The design question is whether the \$/WAR rate should be estimated on one-year contracts only (length=1) to isolate the pure spot price of production, with multi-year term risk handled entirely in the discount rate.

Q2. Is the intercept drift a cap-inflation residual or a genuine market shift?

Year FE significance is driven by a downward drift in intercepts over the sample. The most likely explanation is that below-floor salaries do not scale proportionally with cap growth, so the cap% denominator slightly understates true inflation for replacement-level talent. A robustness check — expressing cost as deviation from the era-mean cap% — would test whether the drift disappears.

Q3. Do goalies belong in the same regression as skaters?

The current specification pools skaters and goalies. Goalie WAR from Bacon's GSAX model is on a different scale and more volatile than skater WAR. A few Vezina-calibre contracts may be pulling the intercept. Running skaters-only as a robustness check would confirm whether the pooled specification is appropriate.

Q4. Does the WAR slope exhibit non-linearity at the top of the distribution?

Scatter plots suggest a possible superstar premium where elite players command more cap% per WAR unit than the linear slope predicts. A quadratic WAR term or spline specification would test this. If confirmed, the model systematically undervalues elite players in trade evaluations.

Q5. How should rapid cap growth (2025-26 onward) be handled in terminal value projections?

The cap jumped \$7.5M in 2025-26 to \$95.5M, with further increases projected to ~\$104M (2026-27) and ~\$113.5M (2027-28). RFA terminal values computed from a regression fit on 2015-2026 data will be extrapolating into a rapidly expanding ceiling environment. Whether cap-share framing fully neutralises this or whether an explicit growth adjustment is needed for the outer years of multi-year projections is an open question.

9. Next Steps

Ordered by priority for unblocking the NPV model build and back-test.

1	HIGH	Decide: length-adjusted or spot-price regression for the NPV model
		The controlled regression shows length absorbs substantial variance. The NPV framework discounts future production explicitly. Using a length-inclusive regression and a discount rate both pricing term risk may double-count. Decision required before the \$/WAR rate is embedded in the model. Recommended test: re-run on length=1 contracts only and compare the implied \$/WAR to the full-sample estimate.
2	HIGH	Codify era-specific coefficients
		Implement a (era, market) → (intercept, slope) lookup table in the NPV model. Flat-cap coefficients apply to 2020-2022 contracts; normal-cap elsewhere. Coefficients are already estimated — this is an implementation step.
3	HIGH	Source NTC/NMC clause status
		Clause status is absent from the PuckPedia contract export. The friction-cost deduction (Section 2.2) and the NTC/NMC discount contribution are both blocked. Source via supplementary export or targeted scrape.
4	HIGH	Pull goaltender value and NHLe coefficients from Bacon
		Goaltender WAR is available from Bacon but not yet in the stack. NHLe era-varying coefficients are the other missing input. Both are required before any prospect or draft-pick valuation is possible.
5	MEDIUM	Run skaters-only robustness check
		Re-run UFA and RFA regressions on skaters only. Confirm whether pooling goalies pulls the intercept or changes the slope materially.
6	MEDIUM	Test for WAR non-linearity
		Add a quadratic WAR term to both regressions. If significant, the model systematically undervalues elite players — a material issue for trade evaluation.
7	MEDIUM	Test intercept drift correction
		Re-run with cost as (cap% – era_mean_cap%) to test whether the monotone intercept drift disappears. If it does, the current specification has a residual inflation problem.
8	LOW	Draft updated Section 2.3 terminal value language
		Incorporate the era-specific RFA regression into the paper's terminal value formula. Replace QO cost reference with RFA regression cost. Note the length-adjustment decision once resolved.

*Data: PuckPedia player contract export (May 22, 2026). WAR: Patrick Bacon / TopDownHockey (skaters and goalies, 2014-15 through 2025-26).
Cap ceilings: NHL/NHLPA CBA. Contract start year corrected via: $\text{true_start_year} = \text{contract_end_year} - \text{length} + 1$. All regressions use HC3
heteroskedasticity-robust standard errors. Era definitions: pre-flat 2015–2019; flat-cap 2020–2022; post-flat 2023–2026. Baseline: $\text{cap\%} \sim \text{WAR}$. Full
controls: $\text{cap\%} \sim \text{WAR} + \text{age} + \text{age}^2 + \text{contract length} + \text{extension dummy}$.*